MILITARY SPECIFICATION

GAGE, PRESSURE, DIAL INDICATING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of direct indicting oil pressure gages.

1.2 Classification. Gages shall be furnished in the following type designations and dial markings, as specified (see 6.2):

Type AGU-12/A	Fluorescent dial and pointer markings (Replaces type B23)
Type AGU-13/a	White dial pointer markings.

2. APPLICABLE DOCUMENTS

2.1. Government documents.

2.1.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 6620
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MILITARY

MIL-P-116	Preservation, Method of
MIL-F-6836	Flange, Mounting, Attachable, Aircraft Instrument (ASg)
MIL-L-25142	Luminescent Material, Fluorescent

STANDARDS

FEDERAL

FEDERAL STANDARD	
FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS28055	Flange, Mounting, Attachable, Aircraft Instrument (ASg)
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
MS33649	Bosses Fluid Connection-Internal Straight Thread

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unles otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM D3951 Packaging, Commerical

Industry

ANSI-B40.1 Gage, Pressure Indicating Dial type Elastic Element

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be availale in or through libraries or other informational services).

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2.1.2 Other Government documents, drawing, and publications. The following other Government documents, drawing, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Data. Unless otherwise specified in the contract or order, no data are required by this specification or any of the document listed in section 2 (see 6.2).

3.3 General specification. The requirements of ANSI-B40.1 are applicable as requirements of this specification. Additional requirements shall be as specified herein. Where the requirements of this general specification and this specification conflict, the requirements of this specification shall govern.

3.4 Design and construction.

3.4.1 Case. The dimensions of the case shall conform to figure 1.

3.4.1.1 Boss. The boss shall be in accordance with MS33649-04 with "C" dimension omitted.

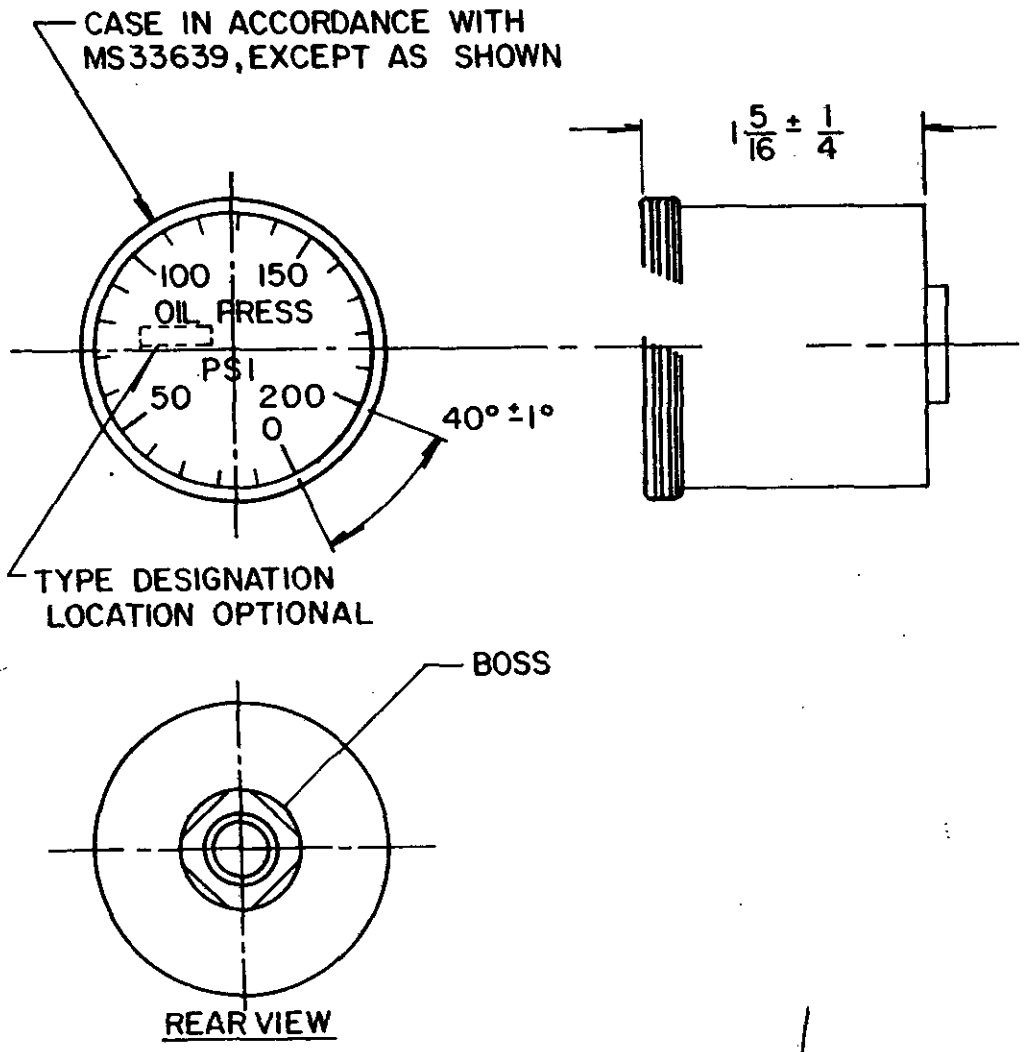
3.4.2 Pointer. The pointer shall be in accordance with MS33585-8, except that the length of the points shall be such that the tip will extend into the scale a distance equal to 1/3 to 2/3 the length of the shortest graduation.

4.3.2.1 Pointer finish. The visible portion of the pointer shall be finished in lusterless white, color No. 595 or fluorescent-luminescent material in accordance with MIL-L-25142.

3.4.3 Dial. The dial shall conform to figure 1. The graduation for 70 pounds per square inch (psi) shall be located on the horizontal centerline at the left side of the dial.

3.4.3.1 Dial markings. Dial markings shall be finished in lusterless white, color No. 37875 of Federal Standard No. 595 or fluorescent-luminescent material in accordance with MIL-L-25142 with dimensions as shown on figure 1. The background of the dial shall be lusterless black, color No. 37038 of Federal Standard No. 595.

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MARKINGS	HEIGHT OR LENGTH	WIDTH OF LINE OR GRADUATION
NUMERALS 0, 50, 100, 150, 200,	5/32	0.025
50-POUND GRADUATIONS	3/16	0.031
10-POUND GRADUATIONS	3/32	0.020
LETTERING "OIL PRESS."	1/8	0.025
LETTERING "PSI"	1/8	0.025

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS ± 1/64, DECIMALS ±0.005

FIGURE 1. Case and dial

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3.5 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.6 Recycled and reclaimed materials. The use of recycle and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 General specification. The quality assurance provisions shall be in accordance with ANSI-B40.1 and as specified herein.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspetion requirements in the specification shall not relieve the contractor of the responsibity of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifcation where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

4.3 Inspection condtions. Unless otherwise specified all inspection shall be performed in accordance with the test condition specified in 4.6 this specification.

4.4 Classification of inspections. The inspection and test specified herein shall be classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.5)

4.4.1 First article test samples. The number of samples and test requirements shall be as specified in ANSI-B40.1.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the following individual and sampling tests. Performance of the gage shall be checked against the error tolerances in tables I and II.

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TABLE I. Scale error and friction error tolerances.

Test pressure	Scale error pressure (±psi)		Friction error pressure (±psi)	
	Room temp	High and low temp	Room temp	High and low temp
0	3.0	4.0	----	-----
50	3.0	6.0	5.0	6.0
100	5.0	6.0	----	-----
150	5.0	8.0	5.0	6.0
200	5.0	8.0	----	-----

TABLE II. Position error, vibration, and overpressure tolerances.

Tests	Test pressure (psi)	Tolerances (±psi)
Position error	100	2.0
Vibration error	100	---
Pointer oscillation	---	4.0
Pointer variation	---	3.0
Overpressure	600	3.0

4.5.1 Individual tests. Each gage submitted for acceptance under contract shall be subjected to individual tests in accordance with ANSI-B40.1.

4.5.2 Sampling tests. Sampling tests shall be made in accordance with ANSI-B40.1.

4.6 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, as specified (see 6.2).

5.1.1. Level A.

5.1.1.1. Cleaning. Gage, Oil Pressure shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2. Drying. Gage, Oil Pressure shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3. Preservation application. Not applicable.

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5.1.1.4. Unit packaging. Unless otherwise specified by the contracting activity, each gage, oil pressure shall be Method III of MIL-P-116. Overbox completed pack in PPP-B-636 container. Apply sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the gage, oil pressure when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each gage, oil pressure shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of gage, oil pressure shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2)

5.2.1. Level A. Gage, oil pressure packaged as specified in 5.1.1. shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container be of uniform shape, size, minimum tare and cube consistant with the protection required.

5.2.2. Level B. Gage, oil pressure packaged as specified in 5.1.1. shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3. Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4. Industrial. The packaged gage, oil pressure shall be packed in accordance with ASTM D3951.

5.3. Marking. In addition to any other markings required by the contract or order (see 6.2.), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The oil pressure gage is intended for use in directly indicating the pressure in the main lubrication system of the aircraft engine.

6.1.1 The oil pressure gage can be used as a replacement of the type B-2.2 by the addition of an instrument mounting bezel conforming to MS28055, in accordance with MIL-F-6836.

6.2 Ordering data. Procurement documents should specify the following:

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(a) Title, number, and date of this specification.
(b) Data requirements (see 3.2).
(c) Pointer and dial finish (see 3.4.2.1 and 3.4.3.1).
(d) Whether or not sampling plan B is to be omitted (see 4.4.1).
(e) Levels of packaging and packing (see 5.1 and 5.2).
(f) Type designation (see 1.2).

6.3 First Article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be (a preproduction sample), (a first production item) (a sample selected from the first _____production items) (or it may be a standard production item from the contractor's current inventory). The first article should consist of (one), (two), etc units (s). The contracting officer should include specific instructions in all procurement instruments, regarding arragnements for examinations, test and approval of the first article."

6.4 Subject term (key word) listing.
INDICATOR
PRESSURE
2 inch
0-200 PSI

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force -99
Navy -AS

User activities:
Army - EL

Preparing activity:
Air Force -71

Project No.
6620-0425

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-8170D	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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